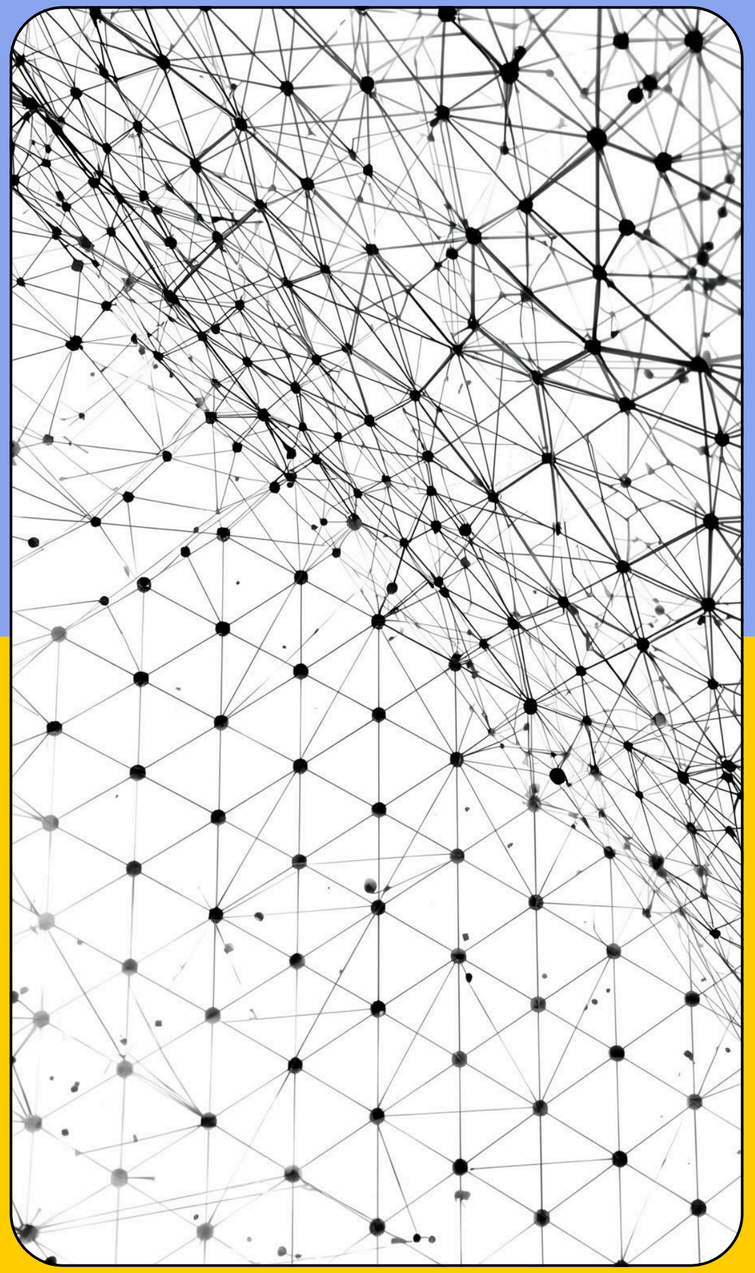
Grid Analysis in Europe

Tools:
Python (Jupyter Notebook & Pandas)
Power BI

Prepared by **Alie Ranti**



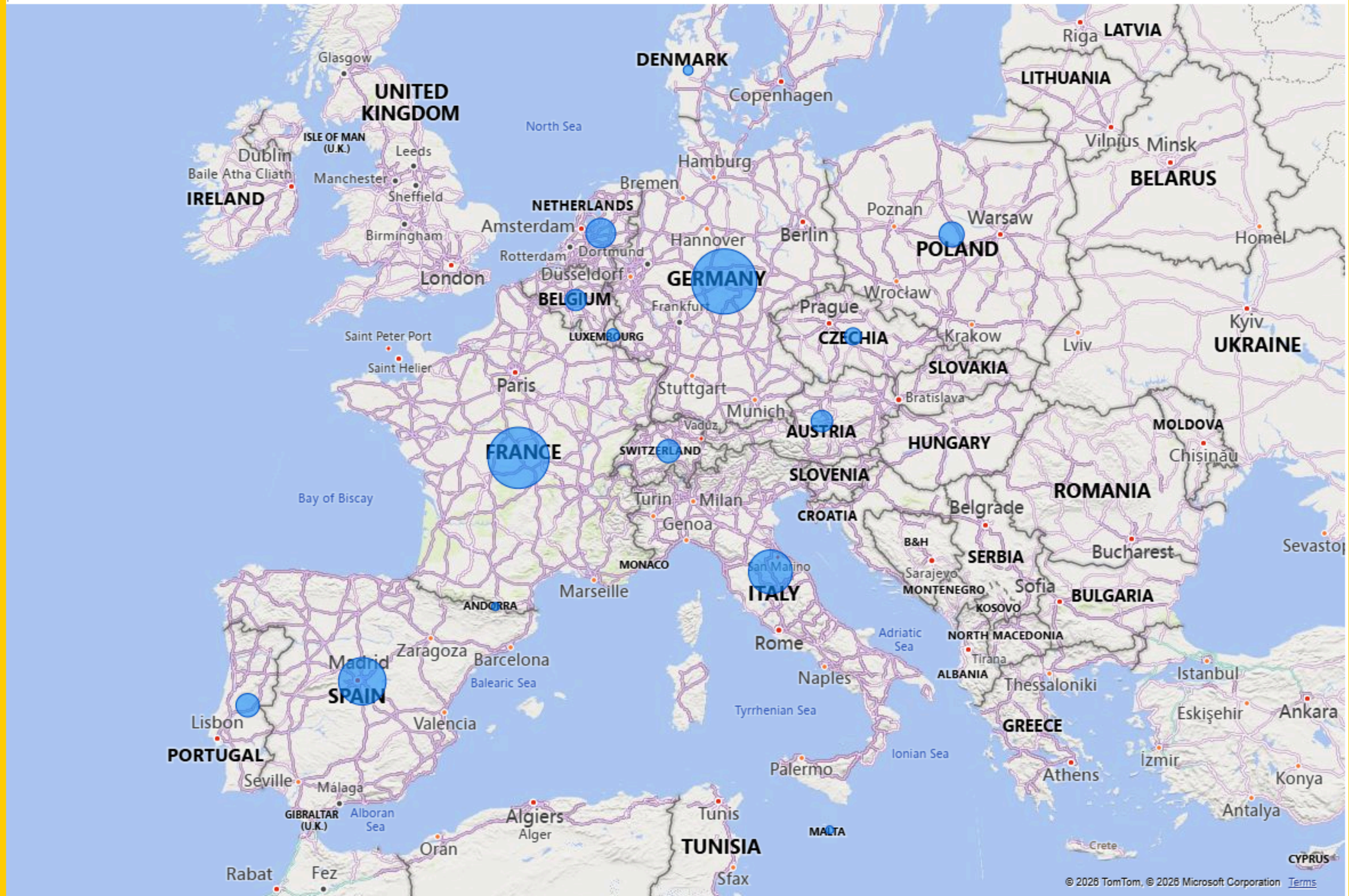
Questions

WHICH EUROPEAN MARKETS SUFFER FROM THE MOST SEVERE GRID CONGESTION?

HOW EXTREME ARE THE PRICE SWINGS WITHIN EACH MARKET?

WHERE ARE THE HIGHEST-PREMIUM INTERCONNECTION NODES IN OUR HOME MARKET (SPAIN)?

HOW MANY INTERCONNECTION NODES ARE IN EACH GRID?



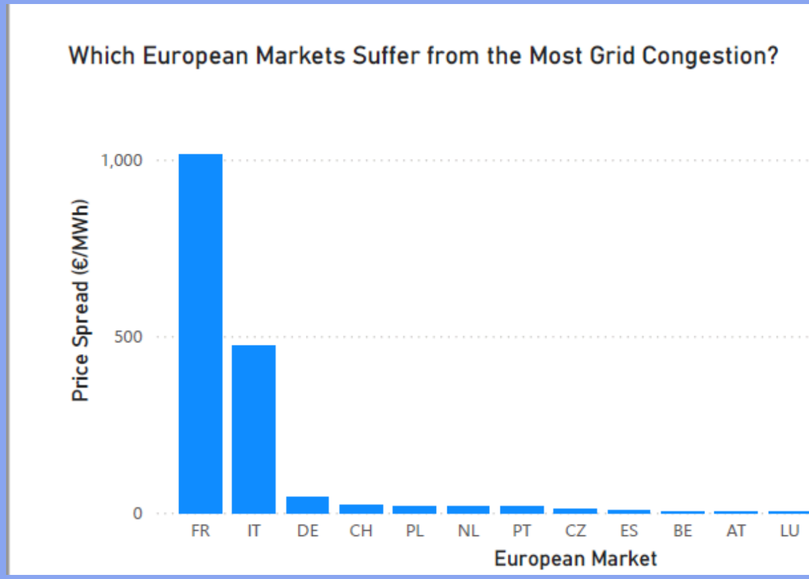
Analyzing Intra-Country Price Spreads

Calculate Intra-Country Congestion

```python
# Group by country and calculate the stats
country_stats = df.groupby('country').agg(
    Avg_Price=('price', 'mean'),
    Min_Price=('price', 'min'),
    Max_Price=('price', 'max'),
    Node_Count=('name', 'count')
).reset_index()

# Calculate the "Price Spread" (Max minus Min)
country_stats['Price_Spread'] = country_stats['Max_Price'] - country_stats['Min_Price']

# Sort the table to see which country has the worst grid congestion at the top
country_stats = country_stats.sort_values(by='Price_Spread', ascending=False)

country_stats
```

	country	Avg_Price	Min_Price	Max_Price	Node_Count	Price_Spread
8	FR	107.718503	-22.405906	992.762078	1729	1015.167984
9	IT	60.152213	38.741022	511.130730	806	472.389709
5	DE	66.350872	28.990119	72.858441	1985	43.868323
3	CH	63.443435	50.904773	71.065604	160	20.160831
13	PL	66.258697	54.980973	72.236505	196	17.255532
12	NL	67.558402	52.992612	69.720008	308	16.727396
14	PT	57.618661	41.732449	57.769723	164	16.037273
4	CZ	66.291927	54.979082	66.488507	64	11.509425
7	ES	57.689369	56.180588	64.151993	965	7.971405
2	BE	70.000892	69.187993	71.471982	125	2.283989
1	AT	66.515032	66.195588	66.872268	128	0.676680
10	LU	69.279848	69.138387	69.398267	21	0.259880
6	DK	66.470875	66.470874	66.470877	6	0.000003
0	AD	63.890157	63.890157	63.890157	1	0.000000
11	MT	54.946263	54.946263	54.946263	1	0.000000

With a massive price spread of 1015, negative minimums, and peaks nearly hitting 1000, **France's** grid is highly congested and volatile. This is the perfect market to build Battery Energy Storage Systems (BESS).



Storage Goldmine:

FR

Peak Shortage:

IT

Highly Stable

ES & DE

What About Spanish?

While **France** requires a battery strategy, **Spain** requires a spatial optimization strategy.

Total Node Counts:

965

Node with Highest Price:

No.25

Top Node and Bottom Node Price Differences:

€8/MWh

Spain

```
# Filter out only the nodes in Spain (ES)
spain_nodes = df[df['country'] == 'ES'].copy()

# Sort them from most expensive to least expensive
spain_nodes = spain_nodes.sort_values(by='price', ascending=False)

spain_nodes
```

	name	price	country
25	26	64.151993	ES
2767	2768	64.114390	ES
3144	3145	64.114390	ES
2364	2365	64.114390	ES
2768	2769	64.114390	ES
...	...	...	...
828	829	56.553490	ES
2211	2212	56.553490	ES
2212	2213	56.553490	ES
3274	3275	56.553490	ES
6	7	56.180588	ES

965 rows × 3 columns